

# Tutors



Open Web Learning Toolkit

## The Educator Experience



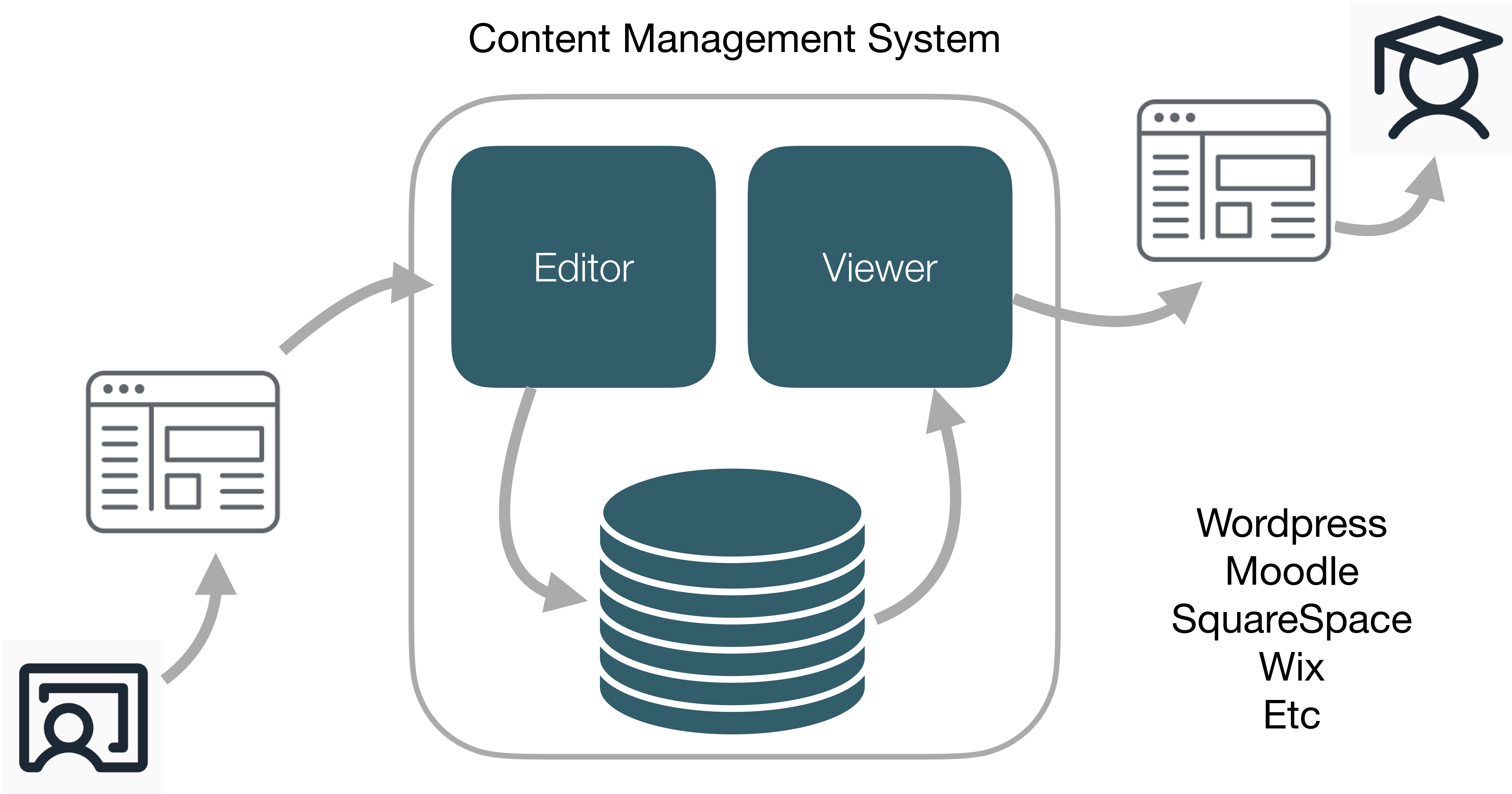
The ***Educator Experience*** prioritises the creation of a **guided paths** through a curriculum via the creation of learning materials that are **autonomous, structurally aligned, composable, auditable, extensible, versioned and independent.**

## Established models for creating & managing web resources

Content  
Management  
System

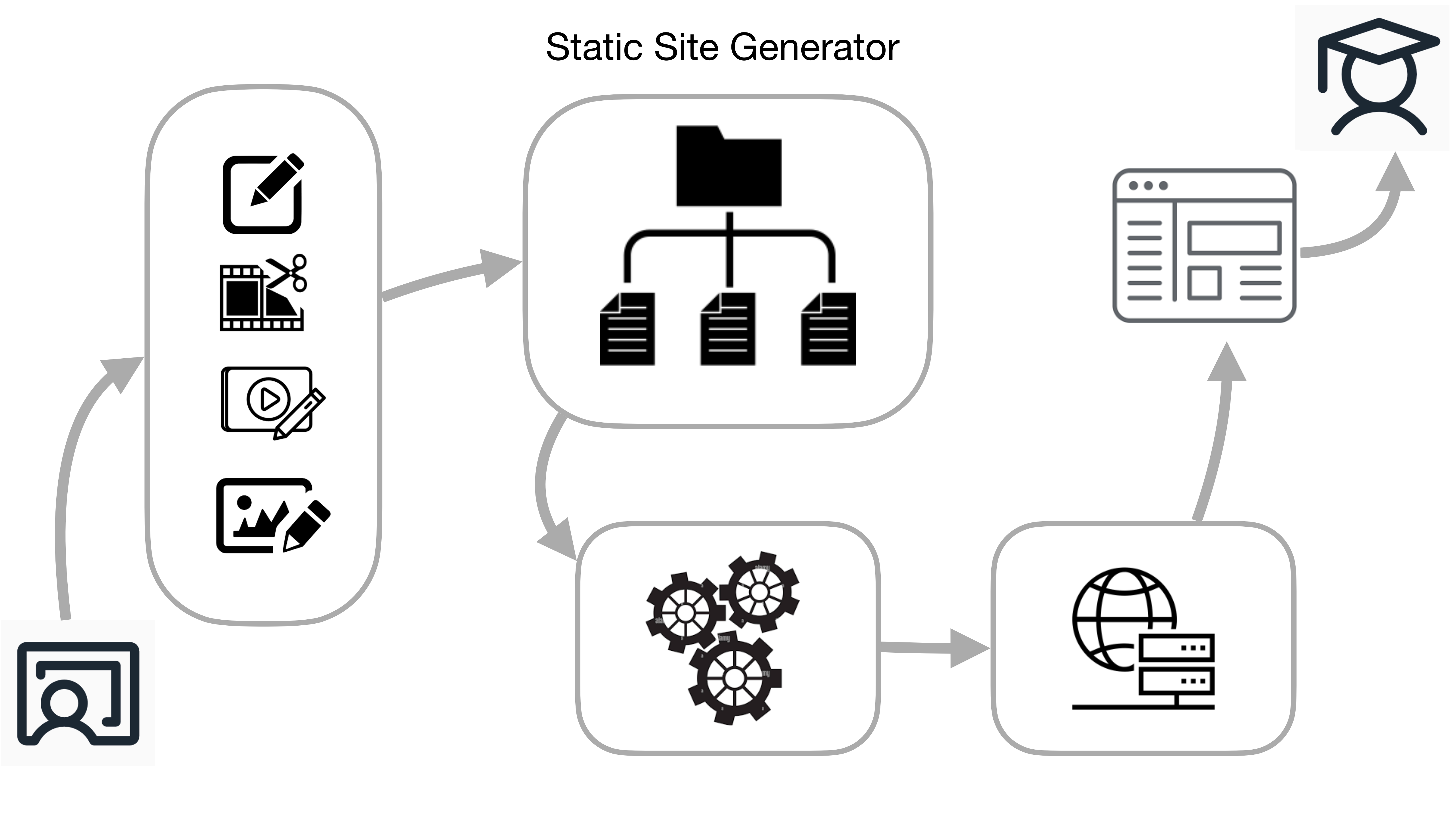
Static  
Site  
Generator

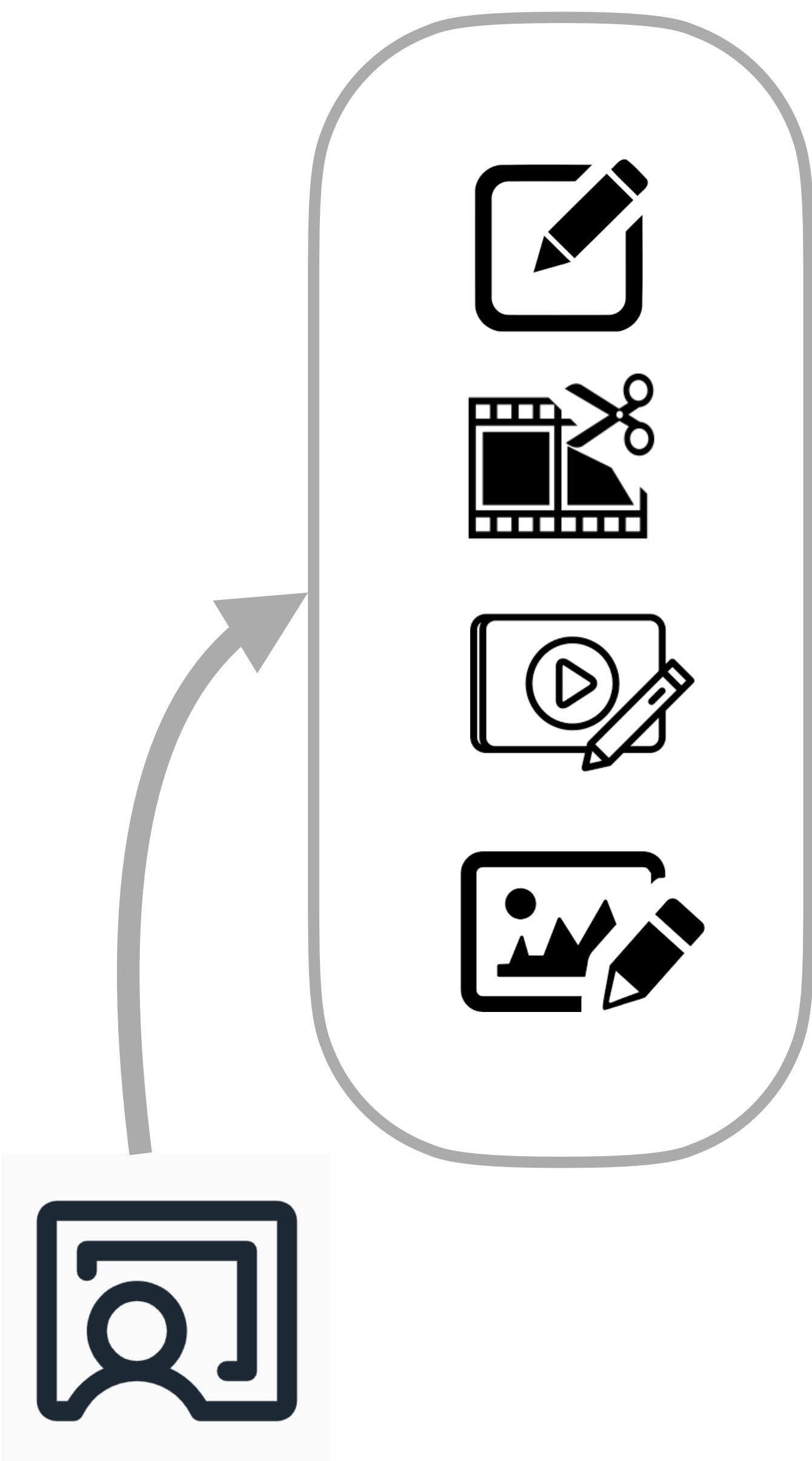
# Content Management System





# Static Site Generator



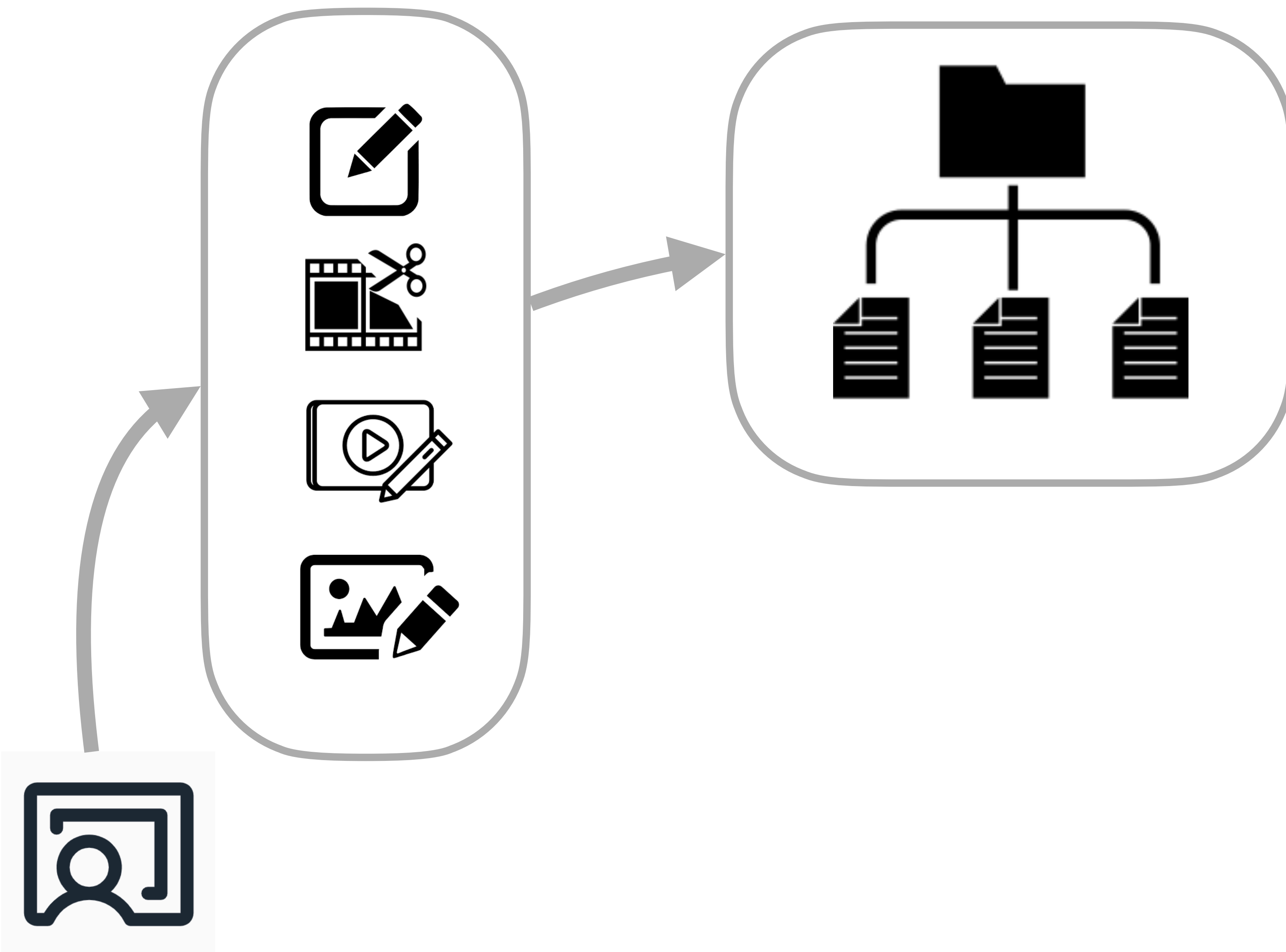


Use whatever tools you like:

- Editors
- Graphic Production
- Icon Sets
- Media

Emphasis on

- Markdown formatted documents
- Latex compatible markdown
- PDFs
- Video Players



Content organised into  
Files & Folders on local HD

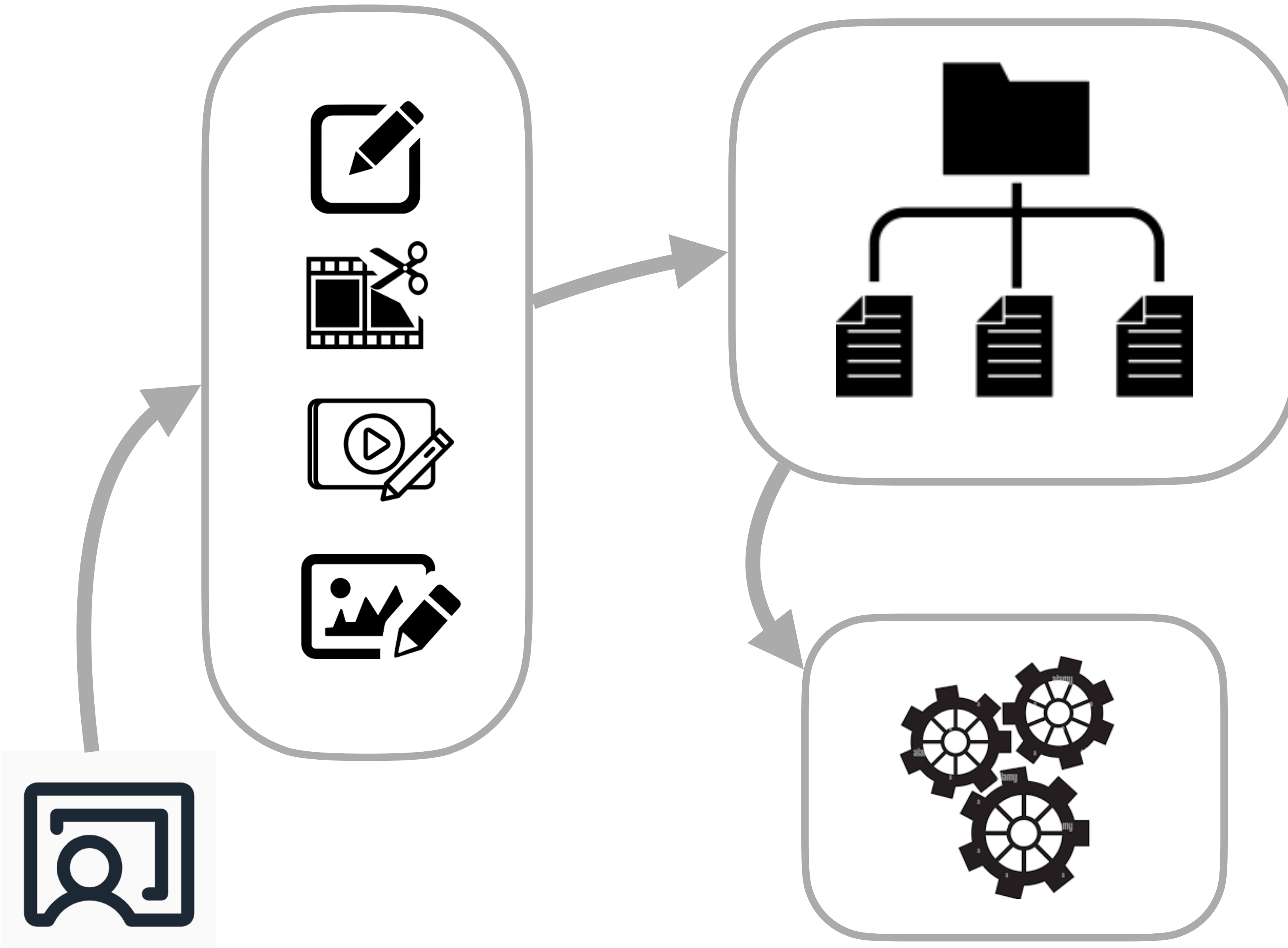
- Nested folder reflects course structure
- Adopt sensible naming conventions & file patterns
- Media & links

Can be versioned and  
stored anywhere

- Github
- Bitbucket
- OneDrive
- Dropbox



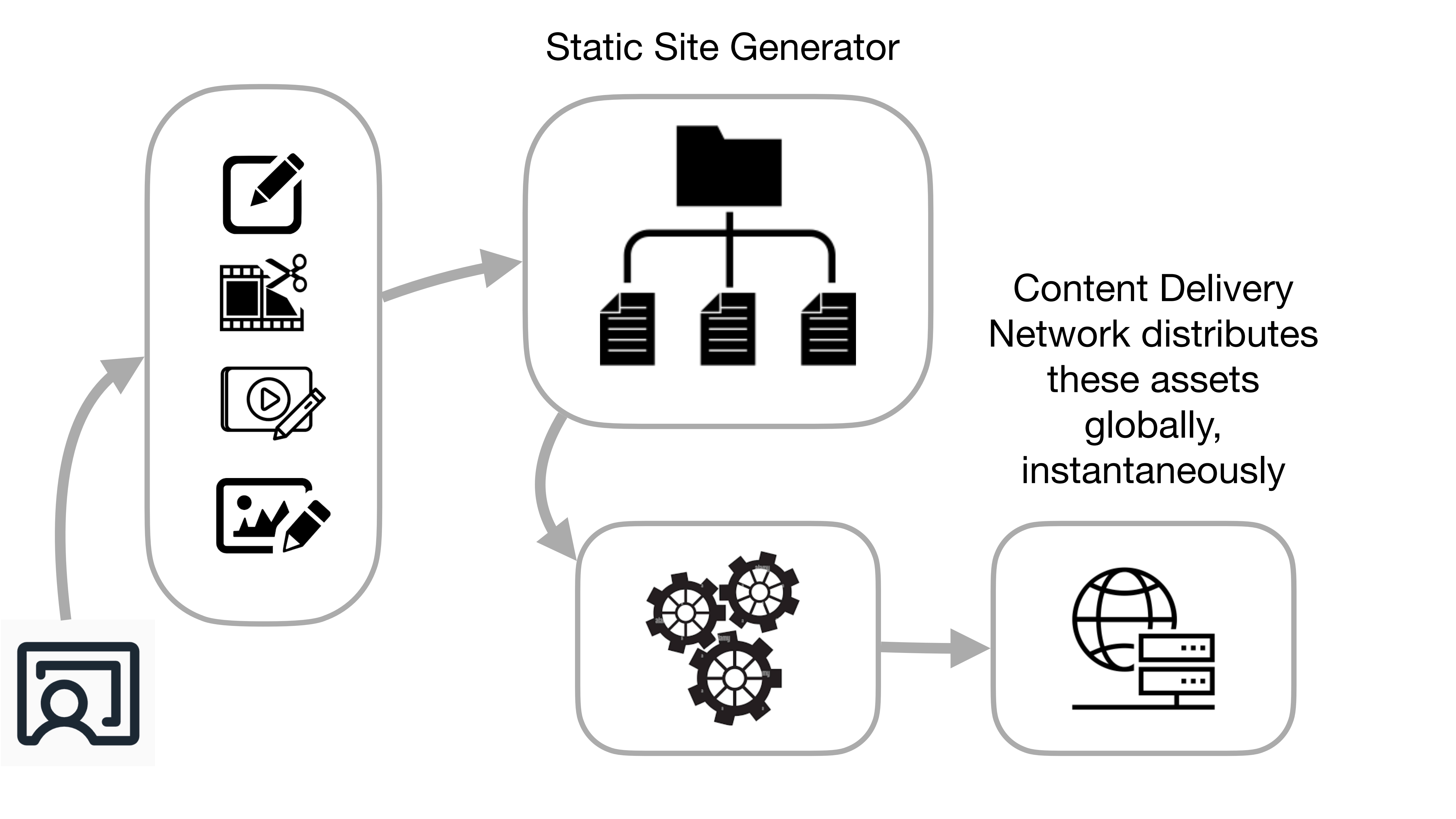
# Static Site Generator



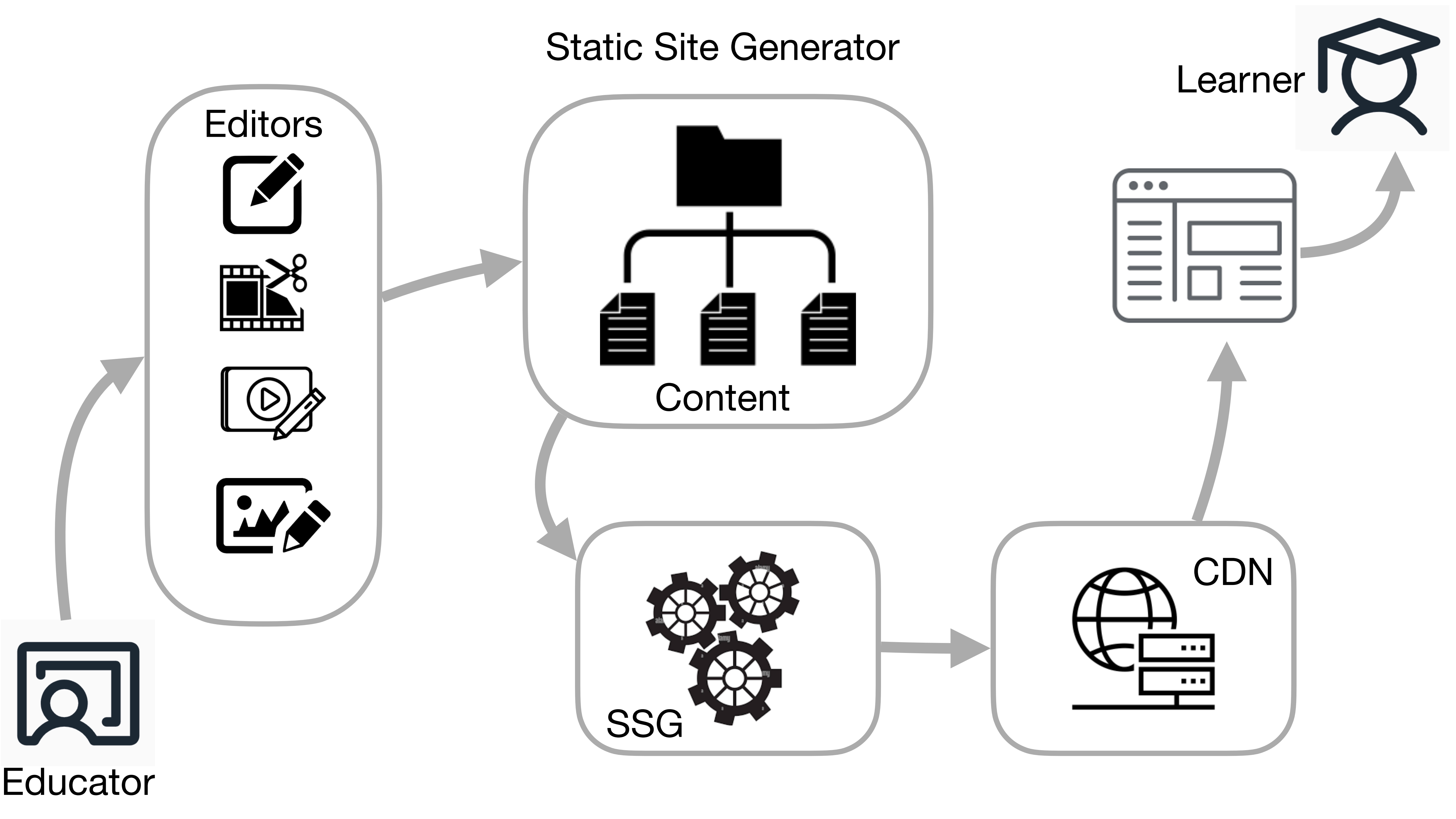
Static Site Generator ingests course structure:

- Produces a set of assets
- Can be directly loaded by a browser
- Can be rendered by a Web Application



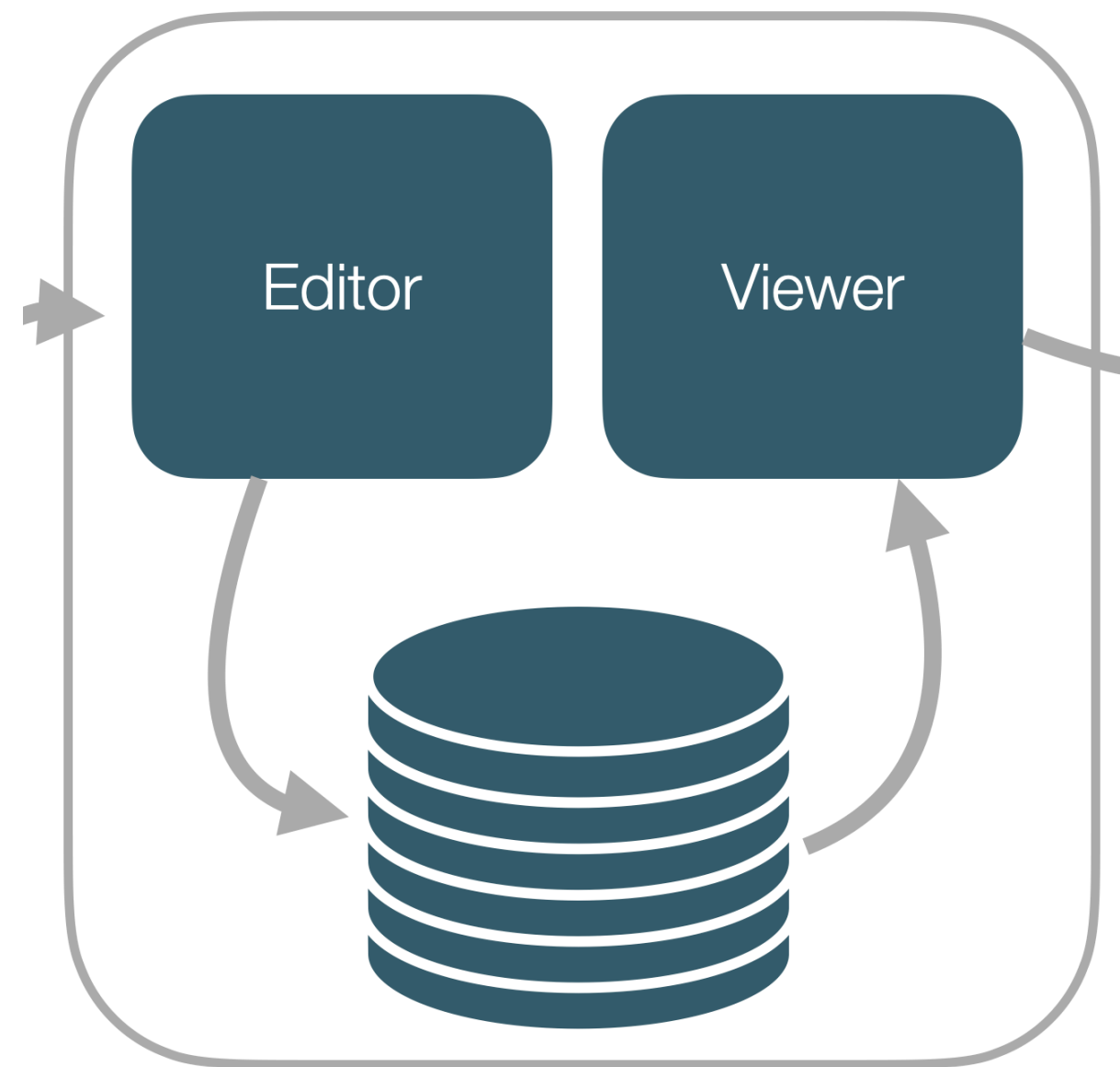






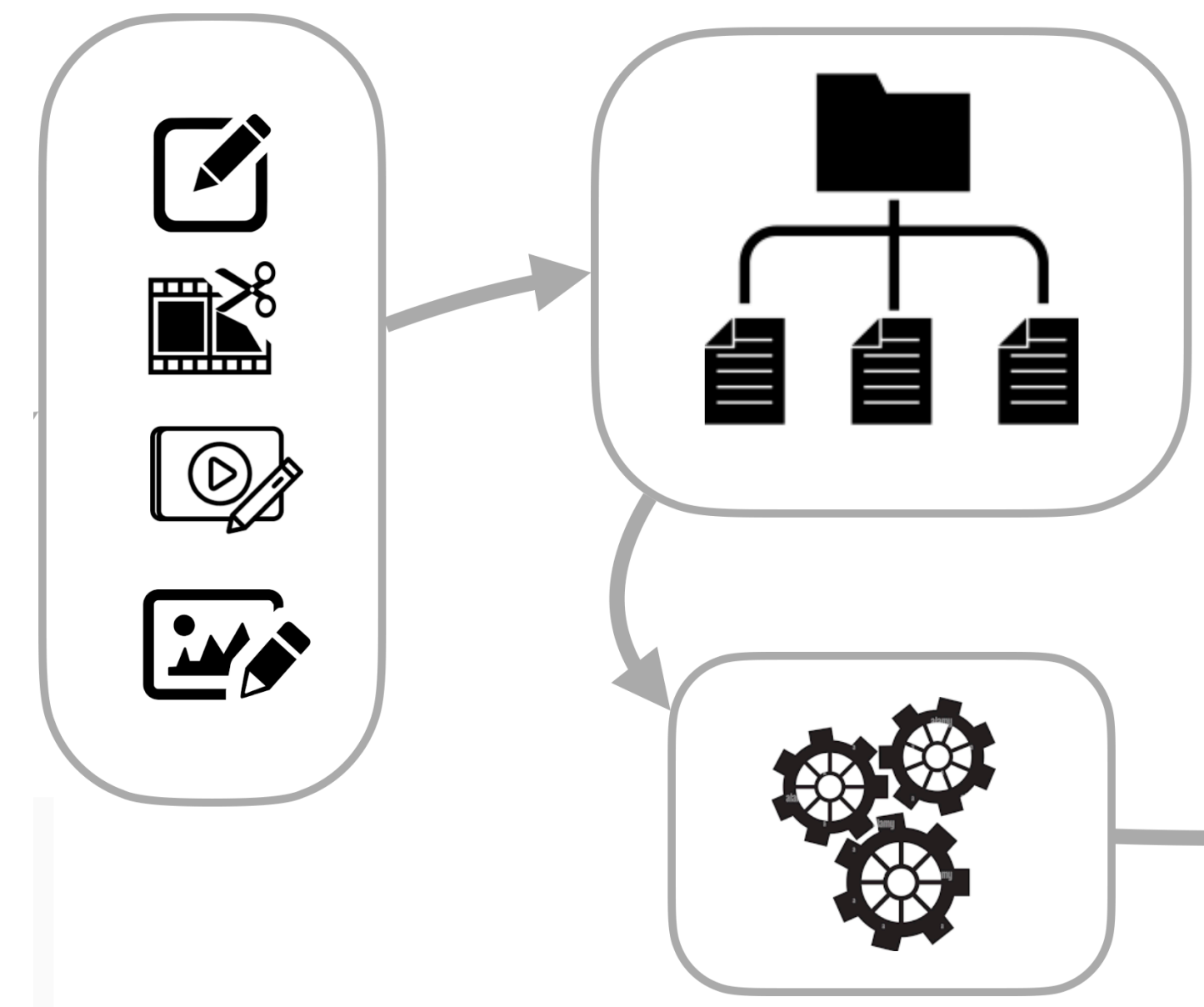
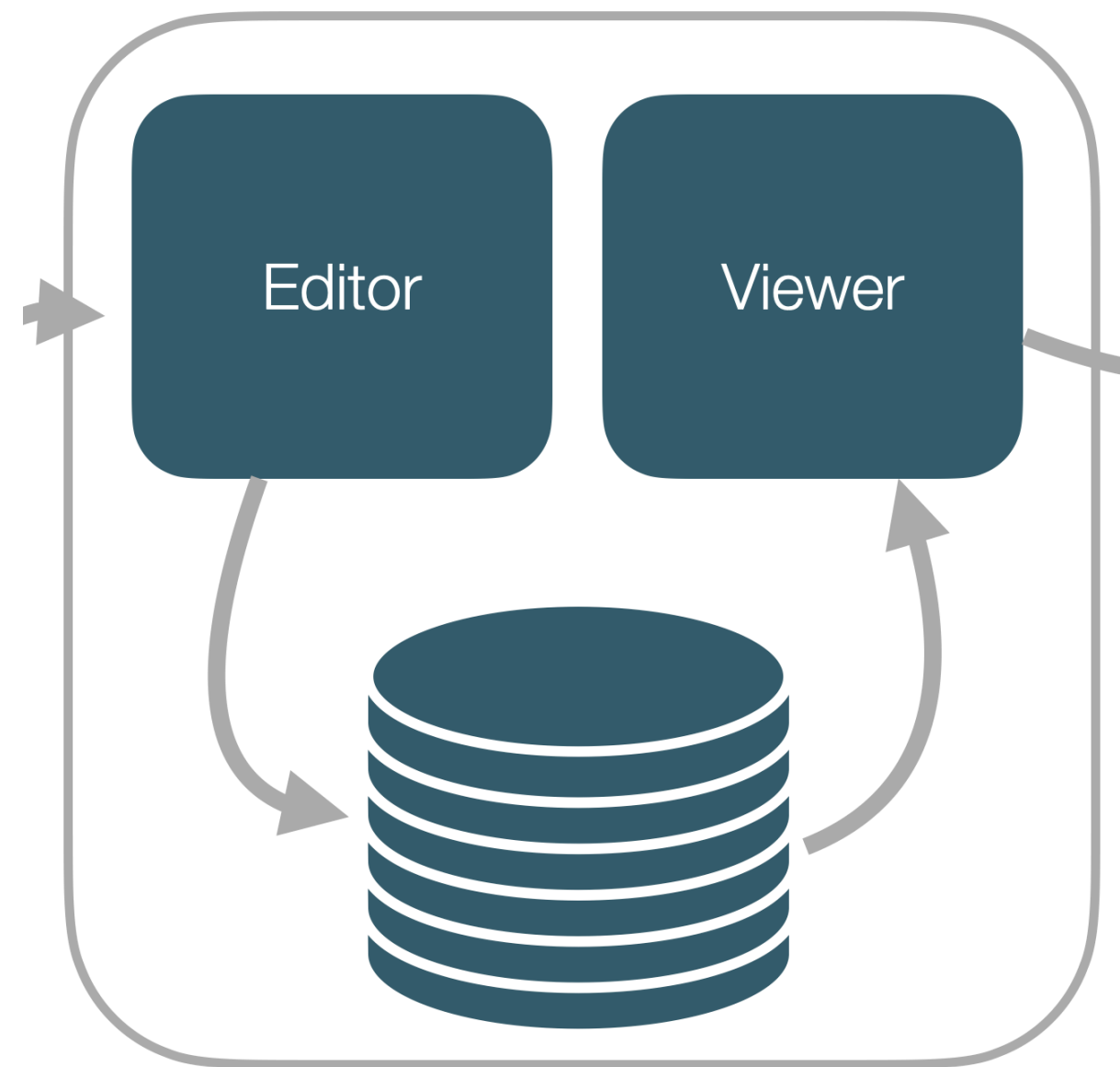
# Content Management Systems

- Course material ‘atomised’ into opaque database
- Dependent on CMS, which must be hosted, secured, patched & maintained
- Courses must be ‘migrated’ each academic year, a painful process
- Highly restricted, often proprietary, design palette



# Content Management Systems

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## Static Site Generators

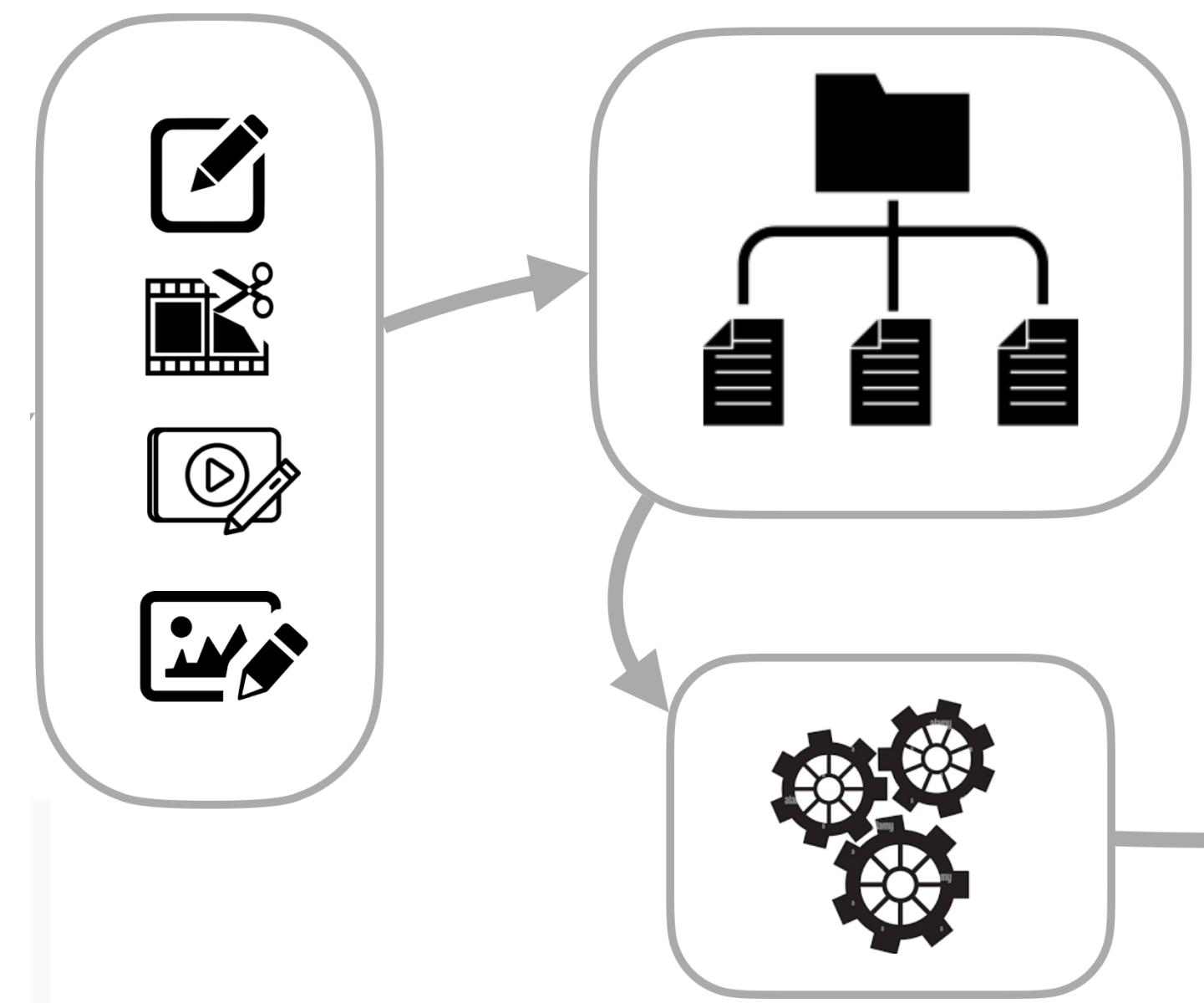
- Course material hand crafted be structurally aligned to course learning paths
- Complete independence, there is no ‘server’ to be maintained
- Course ‘source’ can be managed & versioned (Github)
- New tools can be onboarded as needed



## The Educator Experience



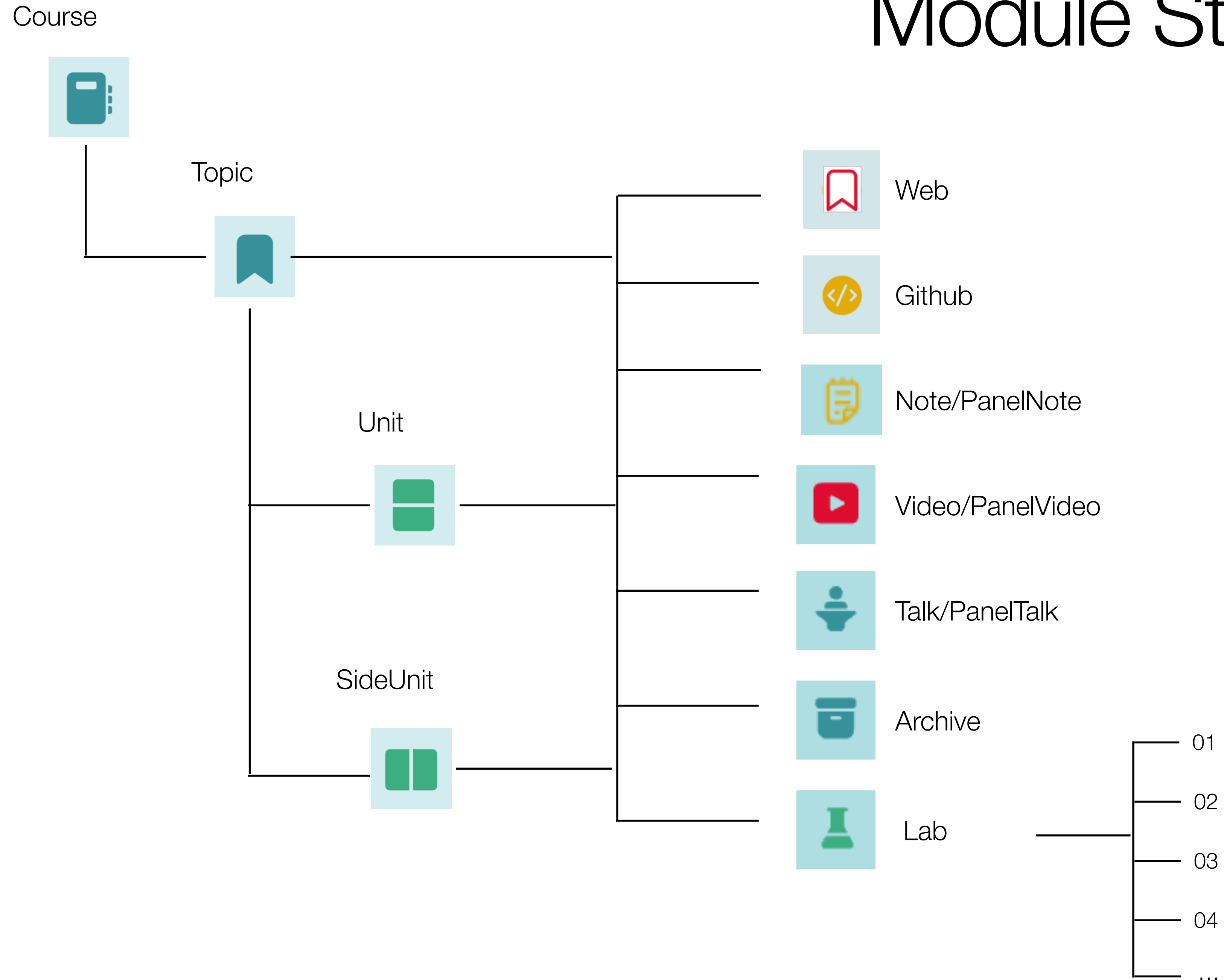
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# Module Structure

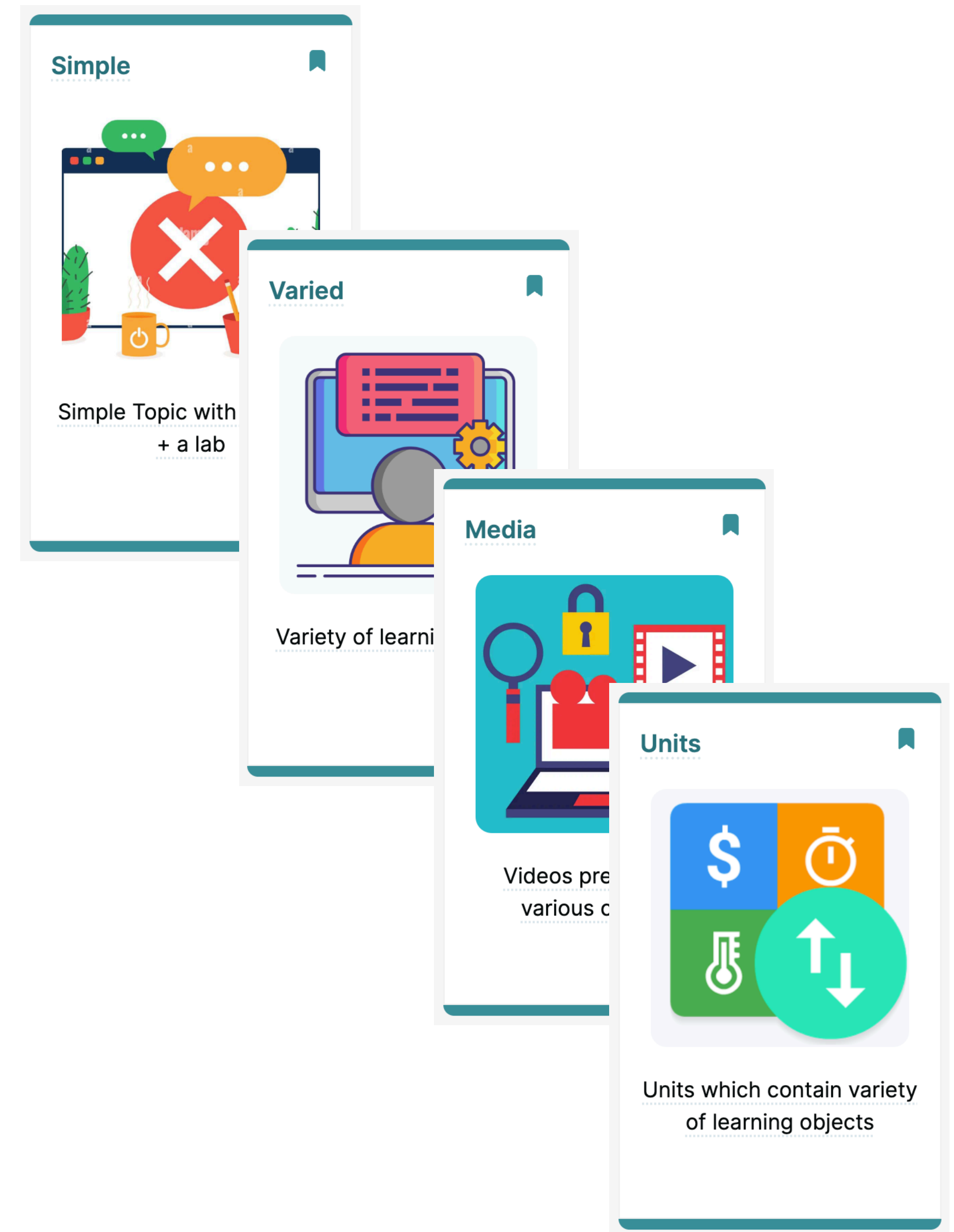




# Topic Structure

<https://reference.tutors.dev>

```
courses/tutors-reference
├── topic-01-simple
│   ├── book-a
│   ├── talk-1
│   ├── talk-2
│   ├── topic.md
│   └── topic.png
├── topic-02-varied
│   ├── archive
│   ├── book-a
│   ├── github-1-template-1
│   ├── note-1
│   ├── note-2
│   ├── talk-1
│   ├── web-1
│   ├── topic.md
│   └── topic.png
├── topic-03-media
│   ├── book-a
│   ├── panelvideo
│   ├── talk-1
│   ├── talk-2
│   ├── topic.md
│   └── topic.png
└── topic-04-units
    ├── unit-1
    ├── unit-2
    ├── topic.md
    └── topic.png
```



# Single Topic Structure



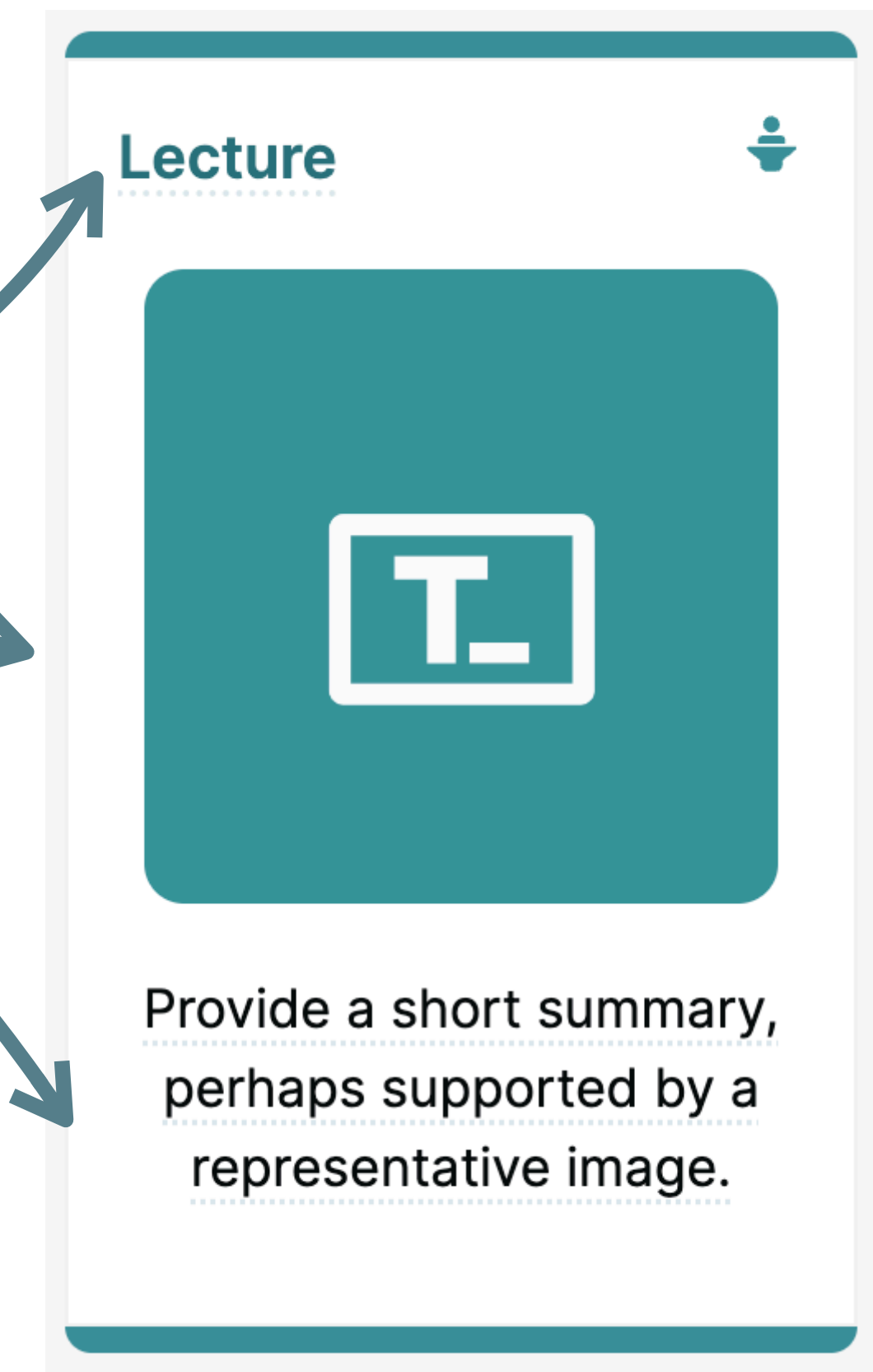


# Talk Structure



all files  
same name

- image
- pdf
- markdown



Lecture

Provide a short summary, perhaps supported by a representative image.

# Lab Structure

book-a

archives

archive.zip

img

01.jpg

02.jpg

03.png

04.png

main.png

00.Lab-01.md

01.01.md

02.02.md

03.03.md

04.04.md

05.05.md

06.Exercises.md

Lab-01: Objectives

01: Text

02: Tables, Lines and Images

03: Links and Code Blocks

04: Images

05: Katex

Exercises: Exercises & Archives

## Links and Code Blocks

You can inert links in bullet points:

- <http://github.com>
- <http://bitbucket.dom>

More Code Block examples

```
publish(path, course) {
  const basePath = '../' + path + '/' + this.folder;
  futils.initEmptyPath(basePath);

  this.resources = this.talks.concat(this.labs);
  futils.copyFileToFolder(this.img, basePath);
  futils.writeFile(basePath + '/index.html', nunjucks.render('topic.html', this));

  futils.writeFile(basePath + '/ajaxlabel.html',
    nunjucks.render('ajaxlabel.html',
      { url: this.url.substring(5) + '/' + this.folder }));
  futils.writeFile(basePath + '/indexmoodle.html', nunjucks.render('indexmoodle.html', this));

  console.log(this.title);
  this.publishTalks(basePath);
  this.publishLabs(basePath, course);
}
```

This was autodetected as javascript. This one is html:

```
<div class="ui segment pushable">
  <div class="ui inverted labeled icon left inline ve
    {% for resource in course.allLabs.resources %}
      {% if resource.title == title %}
        <a class="active item"
href="../../{{resource.parentFolder}}/{{resource.fold
          {{resource.title}}
        </a>
      {% else %}
        <a class="item"
href="../../{{resource.parentFolder}}/{{resource.fold
```

courses > tutors-reference > topic-01-simple > book-a > 03.03.md > # Links and Code Blocks

1 # Links and Code Blocks

2

3 You can inert links in bullet points:

4

5 - <http://github.com>

6 - <http://bitbucket.dom>

7

8

9 More Code Block examples

10

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12 publish(path, course) {

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23 futils.writeFile(basePath + '/indexmoodle.html', nunjucks.render('indexmoodle.html', this));

24

25 con ~/Documents/GitHub/tutors/courses/tutors-

26 thi reference/topic-03-media

27 this.publishLabs(basePath, course);

28 }

29 ~~~

30

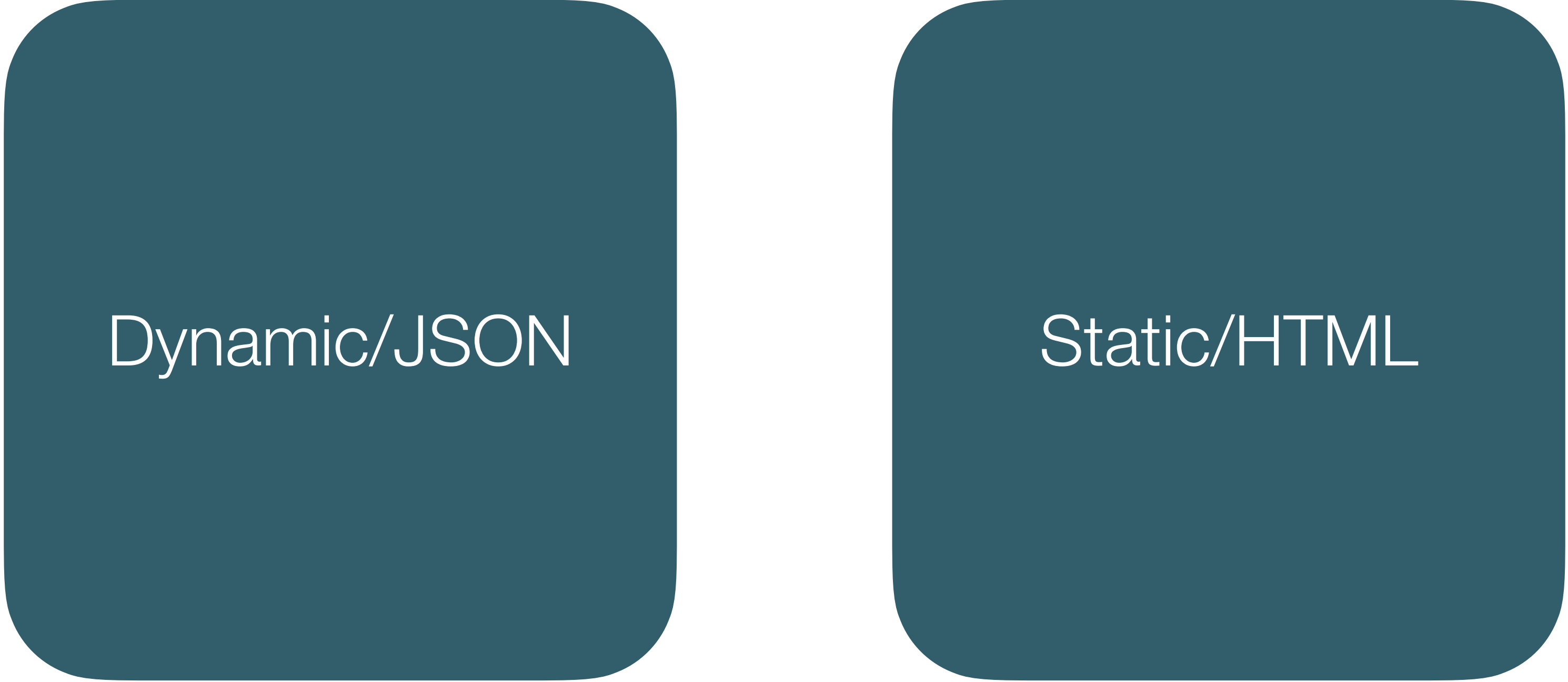
31 This was autodetected as javascript. This one is html:

32

33 ~~~

# Publishing

There are 2 ways to publish a module:



Dynamic/JSON

Static/HTML



# Dynamic/JSON Publishing

- Dynamic Site
- Browser through the Tutors Reader app
- Continuously updated and enhanced
- Included optional analytics or presence features

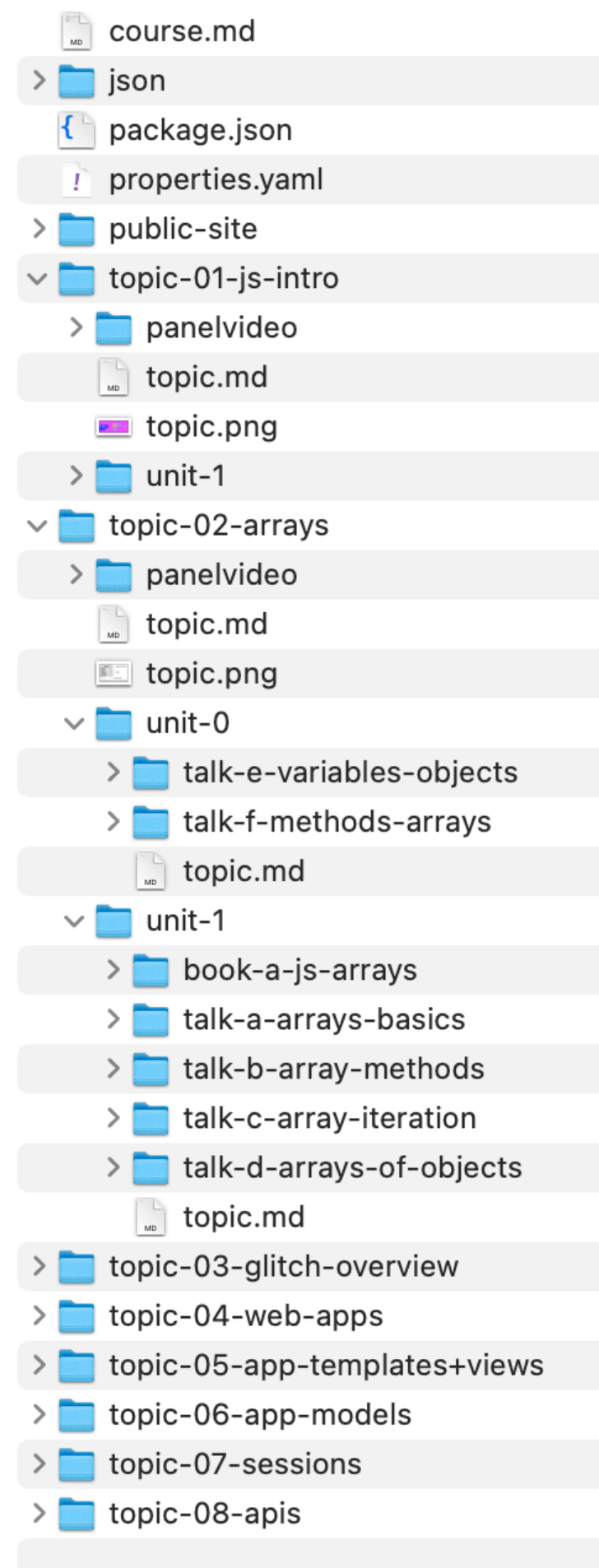
A dark teal rounded square graphic with the text "Dynamic/JSON" centered inside in white.

Dynamic/JSON

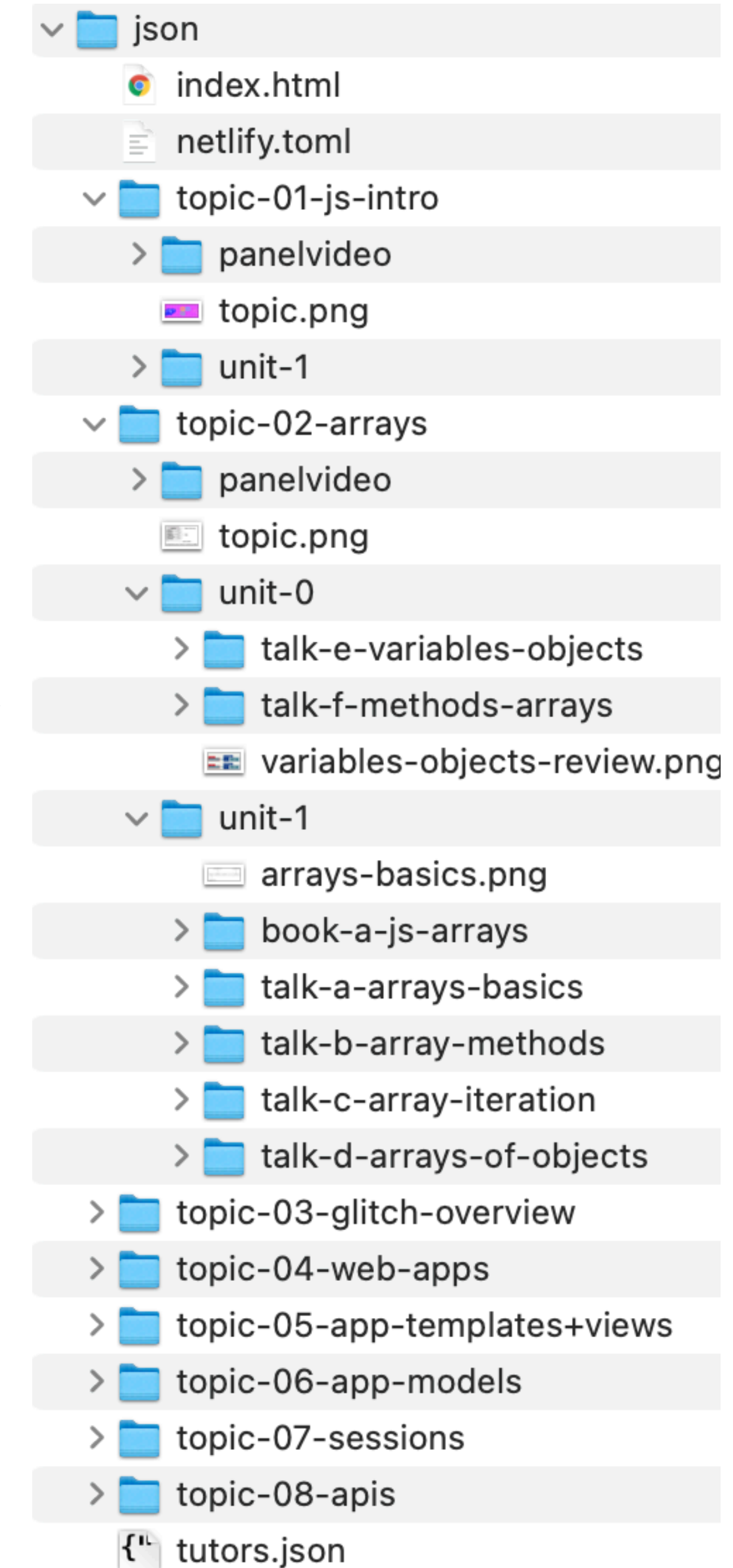


# Dynamic/JSON Publishing

deno run -A jsr:@tutors/tutors

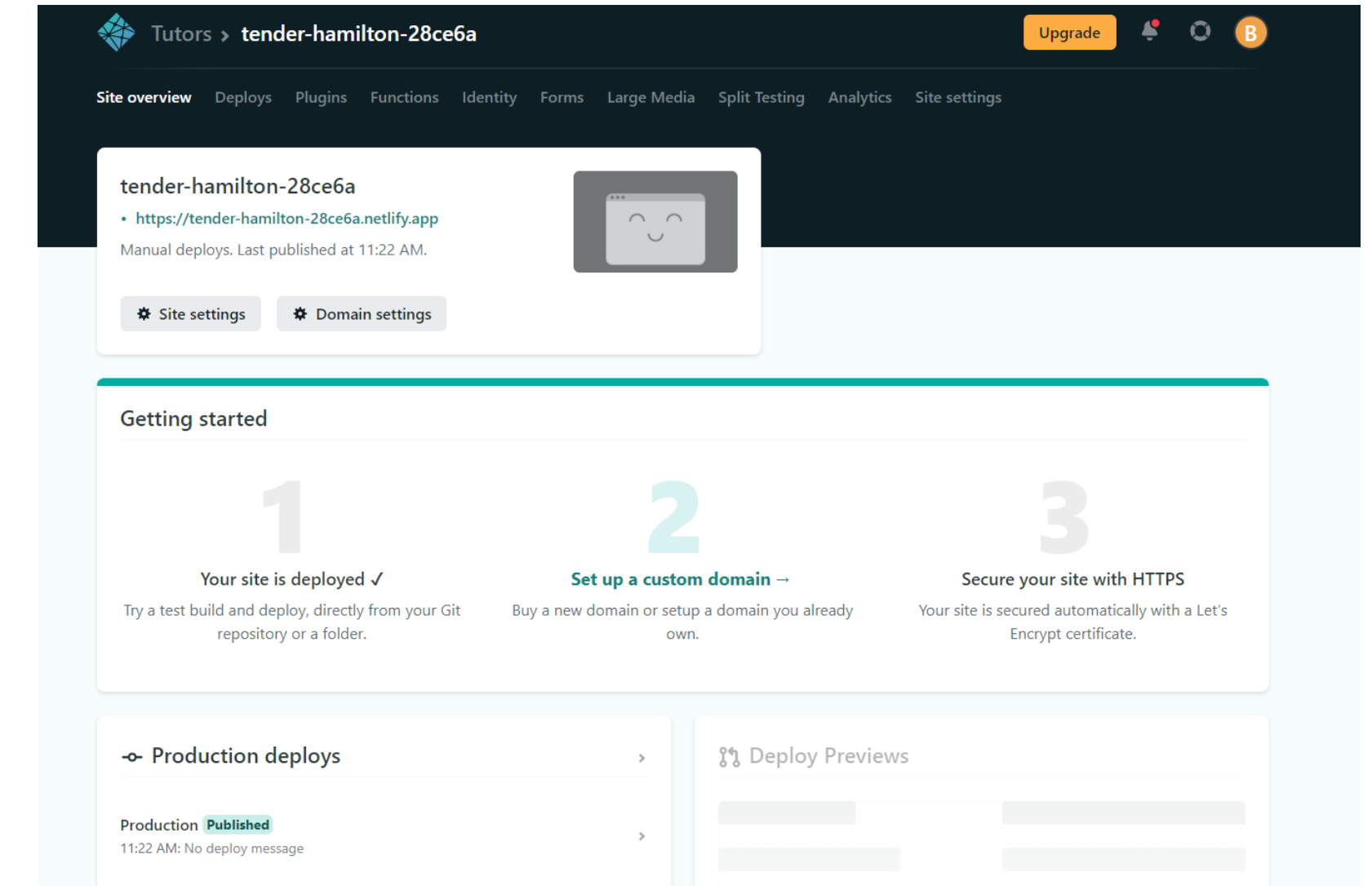
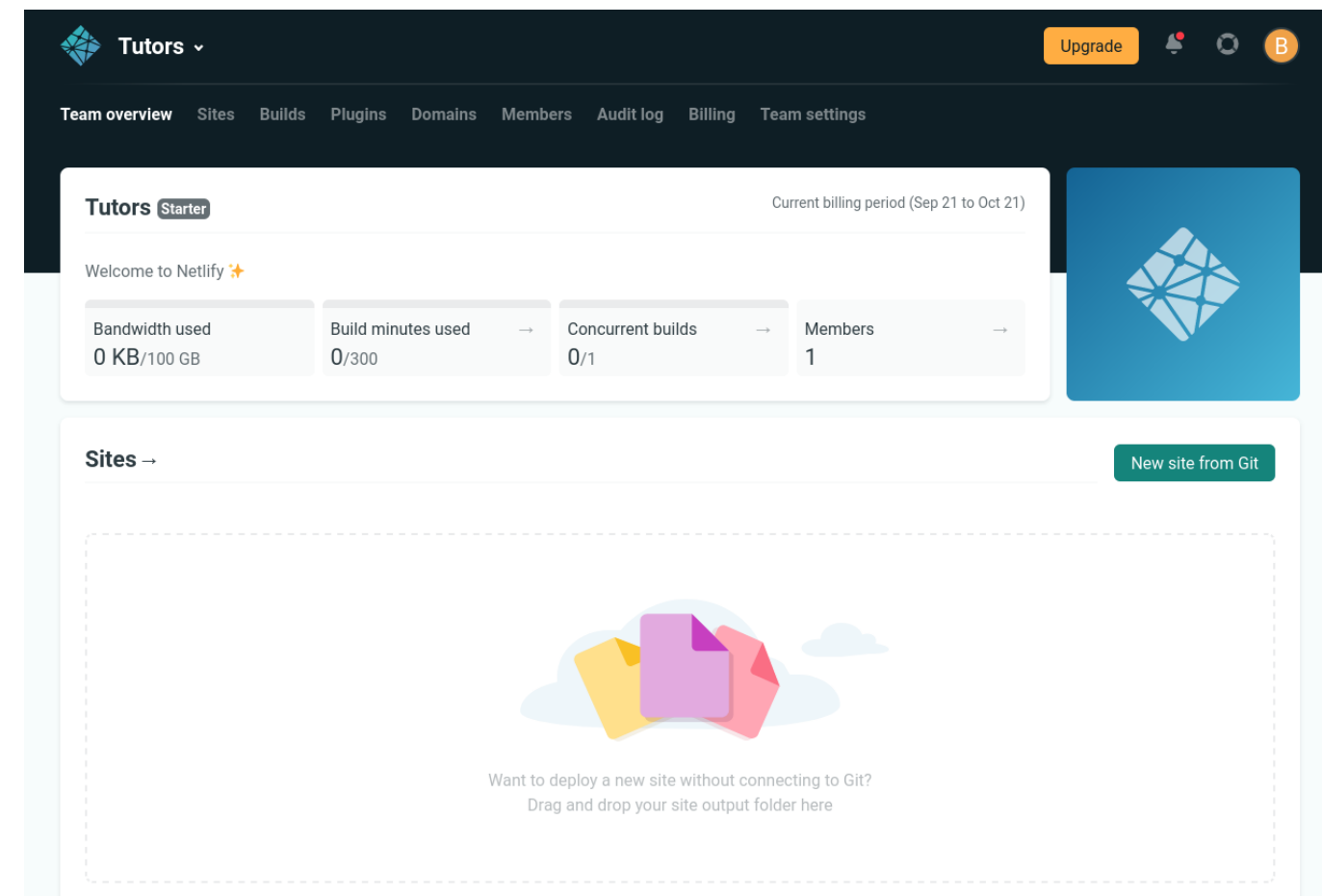
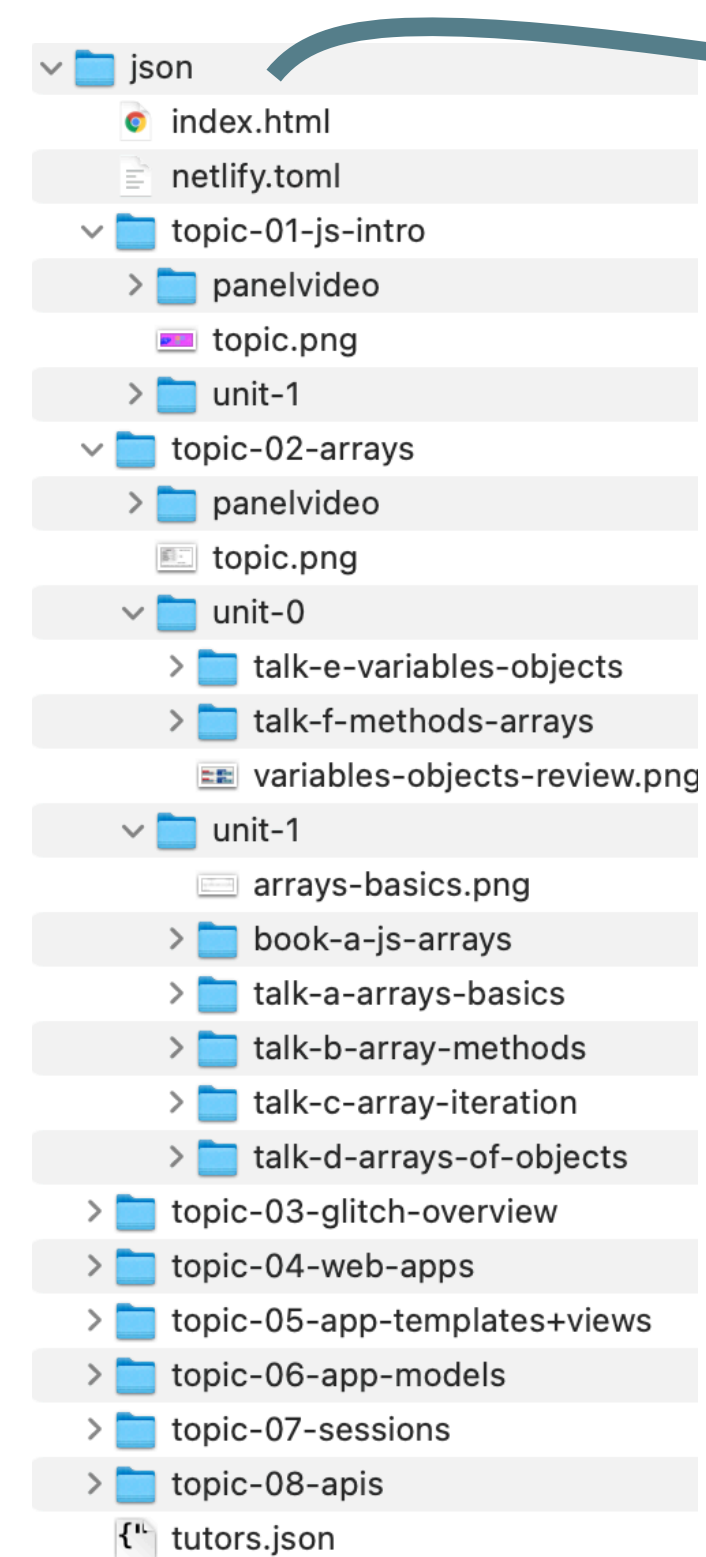


```
-bash
Eamonns-Mac-mini:web-development-2-2022 edeleastar$ npx tutors-json
Static course generator tutors-json 2.2.3 (tutors-lib: ^0.6.3)
:: Web Development II
--> 1: Introducing Javascript (topic)
:: 1: Introducing Javascript
--> Introducing Javascript (unit)
:: Introducing Javascript
--> Module Overview (talk)
--> JS Introduction (talk)
--> Variables & Logic (talk)
--> Const, Let & Objects (talk)
--> Lab-01 JS Intro(lab)
--> 1: Introducing Javascript (panelvideo)
--> 2: Javascript Arrays (topic)
:: 2: Javascript Arrays
--> Javascript Variables, Objects & Methods (unit)
:: Javascript Variables, Objects & Methods
--> Variables & Objects Review (talk)
--> Methods & Arrays Review (talk)
--> Javascript Arrays (unit)
:: Javascript Arrays
--> Arrays: Basics (talk)
--> Array Methods (talk)
--> Array Iteration (talk)
--> Arrays of Objects (talk)
--> Lab-02 JS Arrays(lab)
--> 02: Javascript Arrays (panelvideo)
--> 3: Glitch Overview (topic)
:: 3: Glitch Overview
--> Introducing Glitch (unit)
```



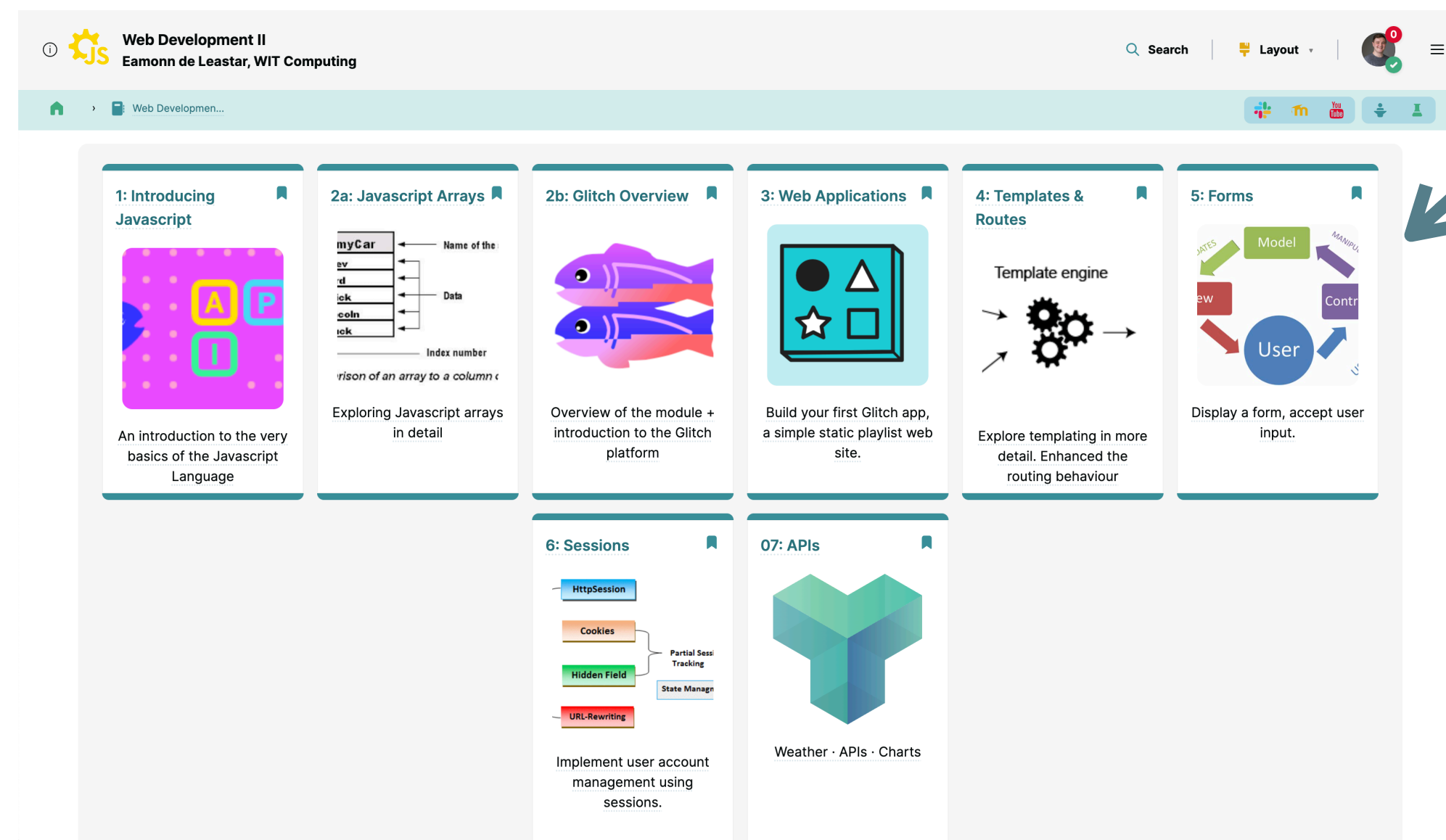
Command generates static site to 'json' folder

# Dynamic/JSON Publishing



1. Netlify GitHub deployment
2. Netlify Drag & Drop

Deploys the Course to be read by the tutors reader





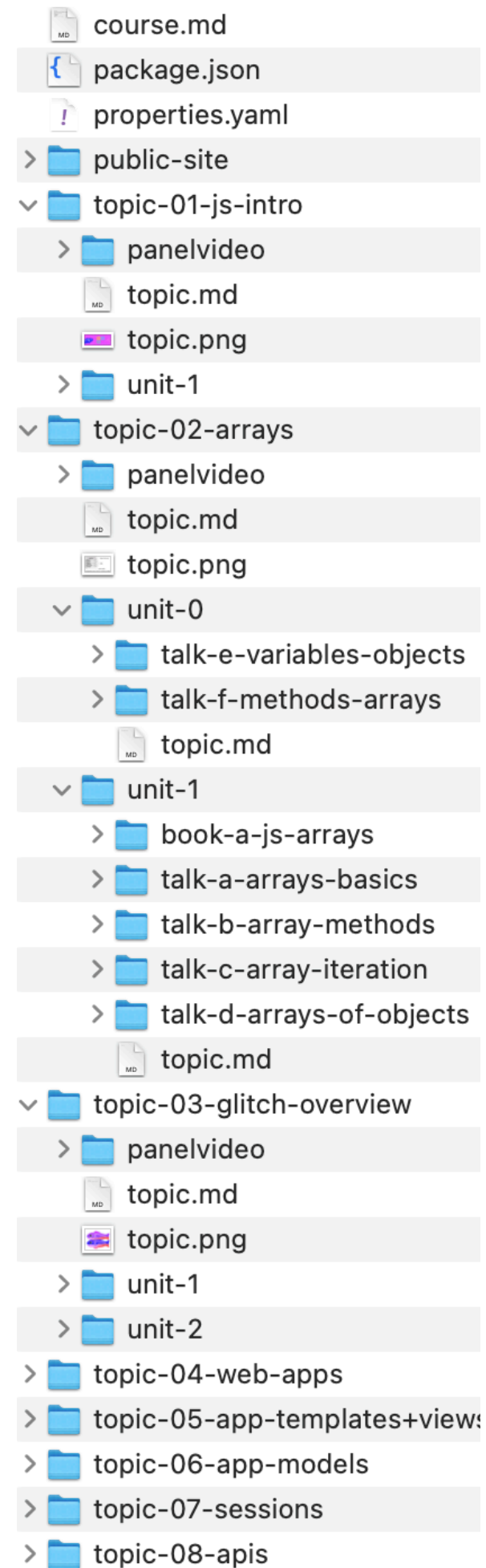
# Static/HTML Publishing



Static/HTML

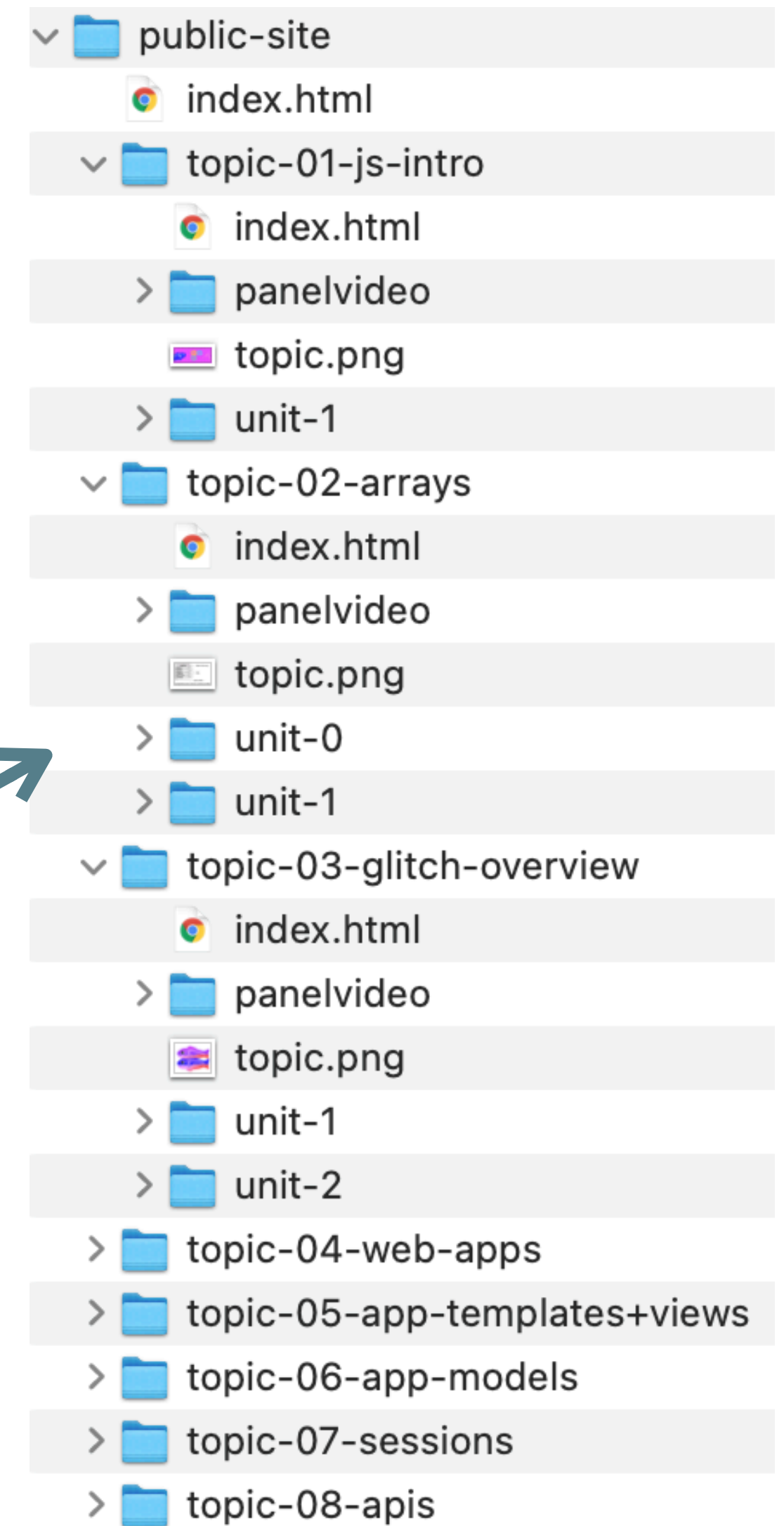
- Static Site
- Can be browsed offline (distribute zipped archive) / uploaded to traditional web host
- Format 'frozen' at time of generation
- No analytics or presence features

# Static/HTML Publishing



deno run -A jsr:@tutors/tutors-lite

```
-bash
Eamonn-Mac-mini:web-development-2-2022 edeleastar$ npx tutors-html
Static course generator tutors-json 0.6.7 (tutors-lib: ^0.6.5)
:: Web Development II
--> 1: Introducing Javascript (topic)
:: 1: Introducing Javascript
--> Introducing Javascript (unit)
:: Introducing Javascript
--> Module Overview (talk)
--> JS Introduction (talk)
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--> Arrays of Objects (talk)
--> Lab-02 JS Arrays(lab)
--> 02: Javascript Arrays (panelvideo)
--> 3: Glitch Overview (topic)
:: 3: Glitch Overview
```



Command generates static site to 'public-site' folder